Total Printed Pages: 02

Roll No.

ECA 101/ EIA 101
Examination – April-May 2026
B.Tech. I Semester EC/EI
Basic Electronics

Time : 03 Hrs

Max. Marks :
Min. Marks :

Note: Total number of questions are 05. All Questions are compulsory. Each Question has 4 pa (a, b, c, d). Part a, b & c are compulsory while Part d has internal Choice. Assume miss data, if any.

Word limit be observed as follows:
Part a – Max 50 words, Part b – Max 50 words,
Part c – Max 100 words and Part d – Max 400 words.
Word limit NOT to be followed for diagram, numerical, derivation.

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Q.1	(a)	What are the majority and minority charge carriers in an p-type semiconductors? explain.	02	2	1
	(b)	How P-N junction is formed? explain.	02	2	1
	(c)	What do you mean by avalanche and Zener breakdown? Compare.	03	1	1
	(d)	Explain the working of a Tunnel diode and draw its V-I characteristics.	05	3	1
		OR			
		Draw and explain the full-wave bridge rectifier.	05	3	4
Q.2	(a)	How many modes are available in Bipolar Junction Transistor ?	02	1	2
	(b)	What is the purpose of higher doping concentration in the emitter region of a Bipolar junction transistor (BJT)?	02	3	2
	(c)	What do you understand by α and β in a transistor? Derive the relationship between them.	03	2	4
	(d)	Explain the circuit diagram and working of a common emitter configuration of an n-p-n transistor.	05	3	2
		OR			
		Draw and discuss the input and output characteristics of a p-n-p transistor.	05	3	2

Q.3	(a)	What are the Z-parameters?	02	1	3
	(b)	Mention the benefits of the h-parameters.	02	1	3
	(c)	The input impedance to a common-emitter transistor amplifier is 1.2 kΩ with β = 120, r_o = 10 kΩ and load resistance is 2.7 kΩ. calculate the value of r_e.	03	3	3
	(d)	Derive the h-parameters of transistor using two port network model.	05	3	3
		OR			
		Draw and discuss the r_e model of a common-base transistor amplifier.	05	3	3
Q.4	(a)	What is the purpose of a multistage amplifier?	02	1	2
	(b)	Differentiate single stage and multistage stage amplifiers.	02	1	2
	(c)	List the different types of coupling used in multistage amplifiers.	03	1	2
	(d)	Draw and discuss the two-stage transformer coupled amplifier.	05	2	2
		OR			
		Two amplifier stages are connected in cascade with 0.5V peak to peak input providing 180V peak-to-peak output. If the voltage gain of first stage is 20 and input to second stage is 18V peak-to-peak determine the overall voltage gain.	05	3	2
Q.5	(a)	What do you mean by the tuned amplifier?	02	1	2
	(b)	Draw the block diagram of a double tuned amplifier.	02	2	2
	(c)	What are the advantages of the double tuned amplifiers.	03	1	2
	(d)	Compare the performances of a Class A and Class B power amplifier.	05	3	2
		OR			
		Discuss the Class C power amplifier with the help of a circuit diagram.	05	3	2